SectraTEK

COURSE OVERVIEW

Five Patterns, One Discipline



A hands-on course in agentic AI for digital forensics. Five AI-agent patterns, taught through real mobile cases, run free in your browser.

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YOUR INSTRUCTOR

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IN ONE LINE

I study how phones get spied on and tampered with, and build AI tools that help examiners find the evidence.

Mobile threat research
Spyware, stalkerware and device tampering.

AI for investigators
Machine learning that speeds up the search.

Teaching
AI for Cybersecurity and Digital Forensics.

START HERE

Who this course is for

You are learning cybersecurity or digital forensics, and you may be new to programming. This course teaches you to use AI agents as careful investigative assistants, step by step.



No deep coding needed

You read and run notebooks, not write them from scratch. Every step is explained in plain language.



Real forensic cases

Each pattern is taught through a synthetic mobile-device case with evidence you reason over.



One habit throughout

Every lab trains the same skill: say only what the evidence supports, and name what it does not.

THE CORE IDEA

From a chatbot to an agent

A plain AI answer

You ask once, it answers once. If the answer overreaches, nothing catches it.

One prompt, one reply.

An AI agent

The model works inside a loop with a goal and a role. It can plan, use tools, check itself, and improve before it answers.

A process, not a single reply.

"Agentic AI" just means wrapping a model in a disciplined process. This course teaches five of those processes.

THE MAP OF THE COURSE

The five agentic patterns



Reflection

Draft, critique, revise.

LAB 1



Tool Use

Call real tools to read the evidence.

LAB 2



ReAct

Reason a step, act, observe, repeat.

LAB 3



Planning

Break a hard case into ordered steps.

LAB 4



Multiagent

Split roles, then reconcile findings.

LAB 5

They build on each other. A Multiagent team is made of ReAct agents that use Tools. Start at Lab 1 and go in order.

THE HABIT THAT MATTERS MOST

Evidence-bounded reasoning

Every lab trains one discipline: separate what the evidence records from what you infer, and never claim more than it supports.



Observation

What an artifact directly records.



Inference

A reasonable interpretation of observations.



Overclaim

A conclusion the evidence does not establish.

THE RECURRING EXAMPLE

A file upload that STARTED is not an upload that COMPLETED. An attempt is not a delivery. Catching that gap, in case after case, is what the whole course is really teaching.

WHAT EVERY LAB LOOKS LIKE

See it, then reuse it

Group activity

Optional: examine the raw evidence as a team and record a human baseline first.

Build it by hand

Run the pattern one step at a time, so every model call is visible.

Run the agent

Do the same workflow through a reusable, packaged class, to see the abstraction.

Evidence check

Answer a few questions that tie each claim back to an artifact.

First you watch the loop work. Then you reuse it. Then you judge it. The judgment stays yours.

NO SETUP, NO COST

Everything runs in your browser

Google Colab

Each lab has an “Open in Colab” button. It opens in your browser. Nothing is installed on your computer.

A free model key

The labs use Groq's free tier. You paste a free key once, it stays hidden, and it is never saved into the notebook.

Self-contained notebooks

Each notebook writes its own case files and carries its own code, so you only ever open one file.

WHAT YOU NEED

A web browser and a free Groq account. No GPU, no admin rights, no Python install. If a model name ever changes, the notebook shows you the current options.

THE SYLLABUS

Five labs, five cases

1	Reflection	Does a leaked customer file prove exfiltration?
2	Tool Use	Was a stolen SUV photographed and listed for sale?
3	ReAct	Did a Signal image send while a phone was unattended?
4	Planning	Reconstruct a phone's timeline during a missing window.
5	Multiagent	Was a patient file transmitted, and is the custody sound?

YOUR FIRST FIVE MINUTES

How to begin

1	**Open the repository**	Go to the course repo and open the Lab 1 folder.
2	**Read the lab README**	It sets up the pattern and the case in plain language.
3	**Click Open in Colab**	The notebook opens in your browser, ready to run.
4	**Paste a free Groq key**	Create one at console.groq.com, paste when asked.
5	**Run it top to bottom**	Follow the Purpose and What you'll see notes on each step.

THE ONE RULE TO REMEMBER

The AI assists. You decide.



It speeds the work

Agents draft, search, and self-check far faster than a person can.



It flags, it does not rule

An agent points at risk and evidence. It does not establish guilt.



A human owns the verdict

A qualified examiner reads the evidence and signs the conclusion, always.

THE HONEST FRAMING

These patterns make an AI a better assistant to an examiner. Not one of them makes the AI the examiner.

SectraTEK

Learn the patterns. Keep the discipline.

By the end you will have built all five agent patterns, each one a better assistant to an examiner, and never the examiner.

